①

$Y = (X+1) \bmod 7 = X \bmod 7 + 1$

yes, they have same distribution

$P(X<4) = 6/7$

② $X, Y \sim Ber(1/2)$ $\quad Z \sim Ber(p) : P(Z=1) = p \;;\; P(Z=0) = 1-p$

$u = x+y \qquad v = |x-y|$

$P(X=1) = 1/2 \quad P(X=0) = 1/2 \quad P(Y=1) = 1/2 \quad P(Y=0) = 1/2$

$u = \{0,1,2\}$

$P(u=0) = 1/4 \quad P(u=2) = 1/4 \quad P(u=1) = 1/2$

$v = \{0,1\}$

$P(v=0) = 1/2 \quad P(v=1) = 1/2$

$u=0 \Rightarrow v=0 \Rightarrow P(u=0, v=0) = 1/4$

$u=1 \Rightarrow v=1 \Rightarrow P(u=1, v=1) = 1/2$

$u=2 \Rightarrow v=0 \Rightarrow P(u=2, v=0) = 1/4$

If they're independent $P(u=u, v=v) = P(u=u) \cdot P(v=v) \;\forall u,v$

③ $\sum_{x=1}^{\infty} c \cdot \frac{2^x}{x!} = 1 \Rightarrow c\left[2 + \frac{2^2}{2!} + \frac{2^3}{3!} + \cdots\right] = 1 \Rightarrow c\left[e^2 - 1\right] = 1 \Rightarrow c = \frac{1}{e^2-1}$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \Rightarrow e^2 = 1 + 2 + \frac{2^2}{2!} + \cdots$

$\sum_{x=1}^{\infty} c \cdot \frac{2^{2x}}{(2x)!} = c\left[\frac{2^2}{2!} + \frac{2^4}{4!} + \cdots\right] = \frac{1}{e^2-1} \cdot \frac{e^2 + e^{-2} - 2}{2} = \frac{(e^2-1)^2}{e^2(e^2-1)\cdot 2} = \frac{(1 - 1/e^2)}{2}$

$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} \Rightarrow e^x + e^{-x} = 2\left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)$

$\frac{e^2 + e^{-2}}{2} - 1 = \frac{2^2}{2!} + \frac{2^4}{4!} + \cdots$

④ Actual : Binomial $n = 500$, $1 \to$ success $= k$, $p = 1/365$

$P(x=k) = {}^{n}C_k \, p^k (1-p)^{n-k} \Rightarrow {}^{500}C_1 \cdot 1/365 \cdot (364/365)^{499}$

approx: poisson : $\lambda = np$ (avg no of successes) $= 500/365$

$P(X=k) = e^{-\lambda}\lambda^k/k!$ $\quad P(X=1) = e^{-500/365}\cdot(500/365)$

⑤

$P(X=n+k \mid X>n) = P(X=k)$

$P(X=k) = p(1-p)^{k-1} \qquad X \sim Geo(p)$

$$P(X=n+k \mid X>n) = \frac{P((X=n+k)\cap(X>n))}{P(X>n)}$$

$$= \frac{P(X=n+k)}{P(X>n)} = \frac{p(1-p)^{n+k-1}}{p(1-p)^n + p(1-p)^{n+2} + \cdots} = \frac{(1-p)^{n+k-1}}{(1-p)^n(1/p)} = p\cdot(1-p)^{k-1} = P(X=k)$$

⑥

a) $P(X=x_1) = 0.1+0.2+0.1 = 0.4$

$P(X=x_2) = 0.15+0.25+0.2 = 0.6$

b) $P(Y=y_1) = 0.1+0.15 = 0.25$

$P(Y=y_2) = 0.2+0.25 = 0.45$

$P(Y=y_3) = 0.1+0.2 = 0.3$

(a) $P(X=x_1 \mid Y=y_2) = 0.2/0.45 = 4/9$

(b) $P(Y=y_3 \mid X=x_2) = 0.2/0.6 = 1/3$

$$P(X=x_1) = \sum_j P(X=x_1 \mid Y=y_j)\cdot P(Y=y_j)$$

$$= \sum_j P(X=x_1)\cap(Y=y_j))$$

$$= 0.1+0.2+0.1 = 0.4$$

⑦ $P(X=k) = {}^nC_k\, p^k(1-p)^{n-k} = {}^nC_k(1/2)^n$

$P(Y=l) = {}^{n+1}C_l\, p^l(1-p)^{n+1-l} = {}^{n+1}C_l(1/2)^{n+1}$

$P(X<Y) = P(n\cdot X < n+1-Y)$

$\downarrow$

$Y-1 < X \Rightarrow Y \le X$

$P(X<Y) = P(X \ge Y) = 1/2$

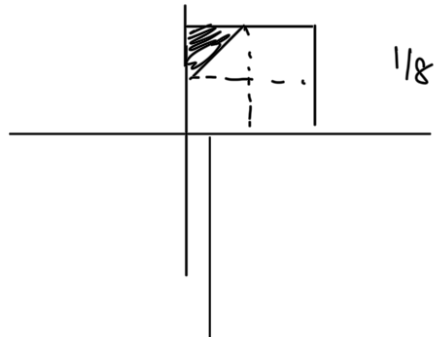
⑧ $X<Y \Rightarrow X,\ Y-X,\ 1-Y$

$Y<X \Rightarrow Y,\ X-Y,\ 1-X$

$X+Y-X > 1-Y \Rightarrow Y > 1/2$

$X+1-Y > Y-X \Rightarrow X+1/2 > Y$

$Y-X+1-Y > X \Rightarrow X < 1/2$



$1/8 \times 2 = 1/4$

(11) angle θ range : $[0, \pi]$

distance of center to nearest line $y \in [0, d/2]$

area of sample space $= (\pi - 0)(d/2 - 0) = \pi d/2$

condition : $y \le l/2 \sin\theta$

$\Rightarrow$ favorable region area $= \int_0^{\pi} l/2 \sin\theta \, d\theta = l$

$\Rightarrow P(\text{crosses}) = \frac{l}{\pi d/2} = 2l/\pi d$

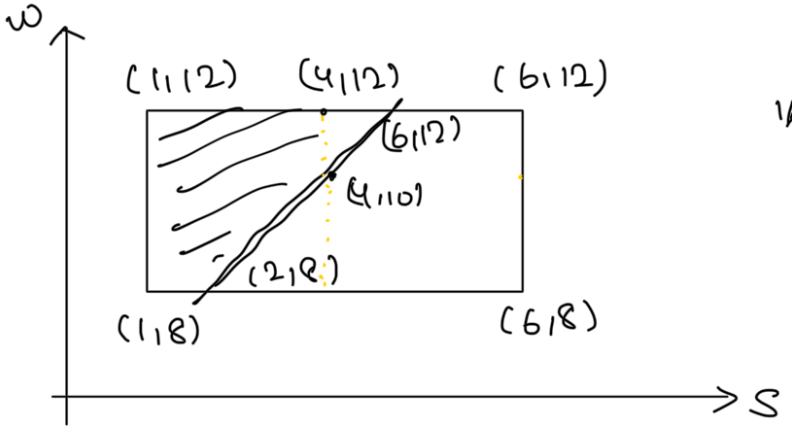
(12) $S \sim$ uniform $(1,6)$ $\quad \omega \sim$ uniform $(8,12)$

$f_S(s) = 1/5 \quad s \in [1,6]$ $\quad f_\omega(\omega) = 1/4 \quad \omega \in [8,12]$

$= 0$ else $\quad = 0$ else

$f_{S,\omega}(s,\omega) = 1/20$

a) 0 b) $P(\omega \le 9) = \frac{9-8}{12-8} = 1/4$ c) $P(\omega - S \ge 6)$



$\frac{1}{2} \times 6 \times 4 = 12/20 = 0.6$

d) $P(\text{enough sleep}) = 3/5$; $P(S \ge 4 \cap \omega - S \ge 6) \Rightarrow \frac{2}{20} \cdot \frac{7}{10} = 7/100$

(9) $F_M(x) = P(M \le x) = P(x_1 \le x) \cdot P(x_2 \le x) \ldots = (F(x))^n$

$F_m(x) = P(m \le x) = 1 - P(m > x)$

$= 1 - P(x_1 > x) \cdot P(x_2 > x) \ldots$

$= 1 - (1 - F(x))^n$

(10) A $\rightarrow$ no of packets arriving from source $P(A = k) = e^{-\lambda} \lambda^k / k!$

$x_1 \rightarrow$ no of packets in buffer after 1st slot $\quad x_1 \in [0,b]$

buffer not full $k<b \Rightarrow e^{-\lambda}\lambda^k/k!$

buffer is full $k=b \Rightarrow A \geq b = 1-P(A<b) = 1-\sum_{j=0}^{b-1} \frac{e^{-\lambda}\lambda^j}{j!}$

$$P(x_1=k) = \begin{cases} \frac{e^{-\lambda}\lambda^k}{k!} & k<b \\ 1-\sum_{j=0}^{b-1} e^{-\lambda}\lambda^j/j! & k \geq b \end{cases}$$

packets left after transmission $= x_1 - \min(x_1, c) = x_2$

$$P(x_2=0) = P(x_1 \leq c) = \sum_{k=0}^{c} e^{-\lambda}\lambda^k/k! \quad j=0$$

$$P(x_2=j) = P(x_1 = j+c) = \frac{e^{-\lambda}\lambda^{j+c}}{(j+c)!} \quad 0<j<b-c$$

$$1-\sum_{k=0}^{b-1} e^{-\lambda}\lambda^k/k! \quad j=b-c$$

$$P(A>b) = 1-P(A \leq b) = 1-\sum_{k=0}^{b} P(A=k) = 1-\sum_{k=0}^{b} e^{-\lambda}\lambda^k/k!$$

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① $x \sim geo(1/2) \quad y \sim geo(1/3)$

$$P(x=k) = 1/2(1/2)^{k-1} = (1/2)^k$$

$$P(y=l) = 1/3\,(2/3)^{l-1} = 2^{l-1}/3^l = 1/2(2/3)^l$$

$$P(x=y) = \sum_{k=1}^{\infty} (1/2)^k \cdot (1/2)(2/3)^k = 1/2 \sum_{k=1}^{\infty} (1/3)^k = 1/2\,\frac{1/3}{2/3} = 1/4$$

$$P(x>y): \sum_{k=1}^{\infty} P(y=k, x>k) = \sum_{k=1}^{\infty} P(y=k)\cdot P(x>k)$$

$$\downarrow \quad 1/2\,(2/3)^k \qquad \downarrow \quad (1-p)^k = (1/2)^k = 1/4$$

② $$P(y=y) = \sum_{x=0}^{\infty} P(y=y/x=x)\,P(x=x)$$

$$\downarrow \quad {}^{x}C_y\, p^y (1-p)^{x-y} \qquad \downarrow \quad e^{-\lambda}\lambda^x/x!$$

$$= \frac{e^{-\lambda}(p\lambda)^y}{y!} \sum_{k=0}^{\infty} [\lambda(1-p)]^k/k! = \frac{e^{-\lambda p}(\lambda p)^y}{y!} \quad y \sim Poisson(\lambda p)$$

$z = x-y$

$$P(z=z) = \sum P(z=z/x=x)\cdot P(x=x) \qquad z=0$$

$$= \sum_{x=z}^{\infty} P(Y=x-z/X=x) \cdot P(X=x) \quad \Rightarrow y = x-z$$

$$= {}_xC_{x-z}\, p^{x-z}(1-p)^z \left[\frac{e^{-\lambda}\lambda^x}{x!}\right] = \text{poisson}(\lambda(1-p))$$

$$P(Y=y, Z=z) = P(Y=y / X=y+z)$$

$$= P(Y=y/X=y+z)\, P(X=y+z)$$

$$= {}_{y+z}C_y\, p^y (1-p)^{y+z-y} \frac{e^{-\lambda}\lambda^{y+z}}{(y+z)!} = \frac{e^{-\lambda}\lambda^{y+z} p^y (1-p)^z}{y!\,z!} = P(Y=y)\,P(Z=z)$$

③ $f_X(x) = 1/4\, x e^{-x/2}$ $\quad f_Y(y) = 1/4\, y e^{-y/2}$ gamma distribution

$$Z = X+Y$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx = \int_{-\infty}^{\infty} 1/4\, x e^{-x/2} \cdot 1/4 (z-x) e^{(x-z)/2} dx$$

$$= \int_0^z 1/16\, x(z-x) e^{-z/2} dx = z^3/6 \cdot 1/16 e^{-z/2}$$

$X \sim par(\alpha) \Rightarrow f_X(x) = \alpha x^{-(\alpha+1)}$ $\quad f_Y(y) = \beta y^{-(\beta+1)}$

$Z = XY$ $\quad U = X \Rightarrow X = U$; $Y = Z/U$ $\quad U \geq 1$; $Z \geq U$

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial u} \end{vmatrix} = |-1/u| = 1/u$$

$$f_{Z,U}(z,u) = f_{XY}(x,y) \cdot |J| = f_{XY}(u, z/u) \cdot 1/u$$

$$Z = XY \quad f_Z(z) = \int_1^z f_X(u) f_Y(z/u) \cdot 1/u\, dx$$

$$= \frac{\alpha\beta}{\beta-\alpha}\left[z^{-(\alpha+1)} - z^{-(\beta+1)}\right]$$

④

A: $[T_A, T_A+0.25]$ $\quad$ B: $[T_B, T_B+0.25]$

$|T_A - T_B| \leq 0.25$ $\quad D = T_A - T_B$

$\mu_D = E[T_A - T_B] = -1$

$\sigma_d^2 = Var(T_A - T_B) = Var(T_A) + Var(T_B) = 0.3\ hr^2$

$D \sim N(-1, 0.3)$

$P(-0.25 \le D \le 0.25)$

Standardize: $Z = \frac{D - \mu_d}{\sigma_d}$

$z_1 = \frac{-0.25 - (-1)}{\sqrt{0.3}} = \frac{0.75}{\sqrt{0.3}}$ $\quad z_2 = \frac{0.25 - (-1)}{\sqrt{0.3}} = \frac{1.25}{\sqrt{0.3}}$

$P(\text{meet}) = \Phi\left(\frac{1.25}{\sqrt{0.3}}\right) - \Phi\left(\frac{0.75}{\sqrt{0.3}}\right)$

(5) $f(x,y) = g(x^2+y^2)$

(a) PD only depends on r, not θ

So knowing R gives no information about θ

$f_{R,\theta}(r,\theta) = f_{XY}(r\cos\theta, r\sin\theta)\,|J|$ $\quad J = \begin{vmatrix} \frac{dx}{dr} & \frac{dx}{d\theta} \\ \frac{dy}{dr} & \frac{dy}{d\theta} \end{vmatrix} = r$

$= r \cdot f_{XY}(r\cos\theta, r\sin\theta)$

$= r \cdot g(r^2)$

↳ function of r

$f_\theta(\theta) = 1/2\pi$ $\quad f_R(r) = 2\pi r\, g(r^2)$ $\quad g(x^2+y^2) = 1/\pi$

(b) $0 \le r \le 1$ $\quad \theta \in [0, 2\pi)$ $\quad g(r^2) = 1/\pi$ (since spherically symmetric)

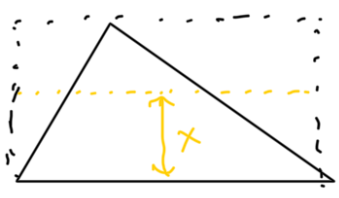
$f_{r,\theta}(r,\theta) = 1/\pi \cdot r$ for $0 < r < 1$

↓ total area

(c) $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ $\quad f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \Rightarrow f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2} = g(x^2+y^2)$

$f_{r,\theta}(r,\theta) = r \cdot \frac{1}{2\pi} e^{-r^2/2}\, g(r^2)$

(6)



$D = 1/2\,bh$

CDF: $F_X(x) = P(X \le x)$

$x \to$ vertical distance from

$\frac{b}{h} = \frac{a}{h-x} \Rightarrow b = \frac{b(h-x)}{h}$ randomly chosen pt to base

$D_{fav} = \frac{1}{2} \times x \times \left(b + \frac{b(h-x)}{h}\right) = \frac{bx}{2}\left[\frac{2h-x}{h}\right] \Rightarrow P = \frac{\frac{bx}{2h}\frac{(2h-x)}{bh}x}{} = \frac{(2hx-x^2)}{h^2}$

$PDF = \frac{d}{dx} F_X(x) = \frac{2h}{h^2} - \frac{2x}{h^2}$ if $x \in [0,h]$

0 else

⑦ $Z = \min(X,Y)$ $X \sim \exp(\lambda)$ $Y \sim \exp(\mu)$

$P(X=x) = \lambda e^{-\lambda x}$ $P(Y=y) = \mu e^{-\mu y}$

$f_{XY}(x,y) = f_X(x) f_Y(y) = \lambda\mu e^{-\lambda x - \mu y}$ $x \geq 0, y \geq 0$

a) $P(X<Y) = \int_0^\infty \int_x^\infty \lambda\mu e^{-\lambda x - \mu y} dy\, dx = \lambda/\lambda+\mu = 1/4$

b) $P(Z>z \text{ and } X<Y) = P(Z>z) \cdot P(X<Y)$ $\forall z>0$

$\Rightarrow P(\min(X,Y)>z) = P(X>z) P(Y>z)$

$= e^{-\lambda z} e^{-\mu z} = e^{-(\lambda+\mu)z}$

$P(Z>z \text{ & } X<Y) = P(z<X<Y)$

$= \int_z^\infty \int_x^\infty (\lambda e^{-\lambda x})(\mu e^{-\mu y})\, dy\, dx =$

$= \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)z}$

$\Rightarrow$ Independent

⑧ use transformation $X = -\ln(U)$

$f_X(x) = e^{-x}$

CDF: $\int_0^x f_X(x)\, dx = 1 - e^{-x} = u$ (inverse transform method)

$\Rightarrow x = -\ln(1-u) \Rightarrow x = -\ln y$

$1-u \in (0,1)$

⑨ $P_{XY}(x,y) = \frac{c}{(x+y-1)(x+y)(x+y+1)}$ $x,y \in \{1,2,3,\ldots\}$

∞ $k-1$

$k = x+y \qquad k \geq 2 \qquad \sum_{k=2}^{\infty}\sum_{x=1} \frac{c}{(k-1)k(k+1)} = \sum_{k=2}^{\infty} \frac{c}{k(k+1)} = c/2 = 1$

$\rightarrow c = 2$

$$P_X(x) = \sum_{y=1}^{\infty} P_{XY}(x,y) = 2\sum_{y=1}^{\infty}\left(\frac{1}{x+y-1} - \frac{2}{x+y} + \frac{1}{x+y+1}\right)$$

$$= 2\left(\frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} + \frac{1}{x+1} - \frac{2}{x+2} + \frac{1}{x+3} + \frac{1}{x+2}\right.$$

$$= 2(1/x - 1/x+1) = \frac{2}{x(x+1)}$$

$$P_Y(y) = \frac{2}{y(y+1)}$$

$$P_Z(z) = \sum_{x=1}^{z-1} \frac{2}{(x+(z-x)-1)(x+(z-x)|(x+(z-x)+1)} = \frac{2}{z(z+1)}$$

$$P(X=Y) = \sum_{k=1}^{\infty} P(X=k, Y=k) = \sum_{k=1}^{\infty} \frac{2}{(2k-1)(2k)(2k+1)}$$

(11) $P(X=2) = 1-p \; ; \; P(X=Y) = p$

(a) CDF of X $\quad F_X(x) = P(X \leq x)$

$$F_X(x) = P(X \leq x / X=Y) \cdot P(X=Y) + P(X \leq x / X=2) \cdot P(X=2)$$

$$= P(Y \leq x) \cdot p + P(2 \leq x) \cdot (1-p)$$

$$= F_Y(x) \cdot p + (1-p) F_2(x) \qquad \text{diff wrt } x$$

(b) CDF $F_X(x) = \int_{-\infty}^{x} f_X(t) \cdot dt$

$$x < 0 \Rightarrow F_X(x) = pe^{\lambda x}$$

$$x \geq 0 \Rightarrow \int_{-\infty}^{0} f_X(t)dt + \int_{0}^{x} f_X(t)\,dt = 1-(1-p)e^{-\lambda x}$$

(10) $P(X=K) = p(1-p)^{K-1}$

$$P(X_i=1, X_j=1) = P(X>i, X>j) = P(X>j) = (1-p)^j$$

$$P(X_i=1, X_j=0) = P(X>i, X<j) = (1-p)^i - (1-p)^j$$

$$P(X_i=0, X_j=1) = P(X<i, X>j) = 0$$

$$P(X_i=0, X_j=0) = P(X<i, X<j) = P(X<i) = 1-(1-p)^i$$

$P(X_i = 1) = P(X > i) = (1-p)^i$
$P(X_j = 1) = P(X > j) = (1-p)^j$ } not independent

---

①

a) $I_{XY} = I(X>Y) = \begin{cases} 1 & \text{if } X > Y \\ 0 & \text{else} \end{cases}$

$a = P(X>Y) = E[I_{XY}]$
$b = E[I_{YZ}]$
$c = E[I_{ZX}]$
$S = I_{XY} + I_{YZ} + I_{ZX} \in \{1, 2\}$

$E[S] \in [1, 2]$
$\Rightarrow 3m \leq 2$
$\Rightarrow m \leq 2/3$

b) $X, Y, Z \sim \text{Unif}(0,1) \Rightarrow a = b = c = 1/2$

②

non-ve
integral is 1

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx = 2$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx = \infty$$

③ k flips head $\Rightarrow p^k(1-p)$
k flips tail $\Rightarrow (1-p)^k p$

$P(X = k) = p^k(1-p) + (1-p)^k \cdot p$

law of total expectation: $E[X] = E[X/A]\,P(A) + E[X/B]\,P($

$E[X / 1^{st} \text{ head}] = 1/1-p$ → until 1st tail, avg flips requir
$E[X / 1^{st} \text{ tail}] = 1/p$ to get 1st tail $= 1/1-p$

$$E[X] = \frac{p}{1-p} + \frac{1-p}{p}$$

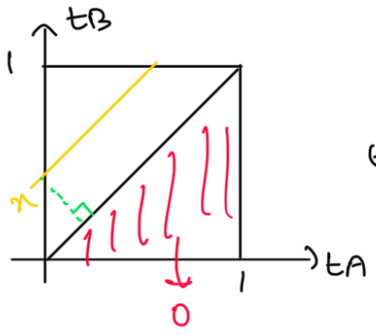
⑥ $E[Y] = \frac{40+33+25+50}{4} = 37$

$E[X] = \frac{1}{N} \Sigma [40^2 + 33^2 + 25^2 + 50^2] = 39.28$

(3) $1-F(x) = P(X>x) = \int_x^\infty f_Y(y)\,dy$

$$E[X] = \int_0^\infty \int_x^\infty f_Y(y)\,dy\,dx = \int_0^\infty \int_0^y f_Y(y)\,dx\,dy = \int_0^\infty y f_Y(y)\,dy$$

(7)



$X = \max(0, t_B - t_A)$

$E[X] = 1/2 \times 0 + \dots$

$X = t_B - t_A$

$f_X(x) = 1-x$

$F_X(x) = P(X \le x)$

$= P(t_B \le t_A + x)$

$= \frac{1}{2} \cdot \frac{1}{2}(1-x)^2$

$1/2 - 1/3 = 1/6$

(8) $V \sim \text{unif}[0,1]$ → true value $E[V] = 1/2$

$f_V(v) = 1$

b → buy

a → sell

uninformed traders:

buy: $u > a$: $E[a-V] = a - 1/2$ $\quad p = 1-a$

sell : $u < b$: $E[V-b] = 1/2 - b$ $\quad p = b$

$$E[\text{uninf}] = (1-a)(a-1/2) + b(1/2-b)$$

$$= -a^2 - b^2 - a/2 + b/2 - 1/2$$

informed:

$$E[\text{inf}] = \int_a^1 (a-v) f_V(v)\,dv + \int_0^b (v-b) f_V(v)\,dv$$

$$= a - 1/2 - a^2/2 - b^2/2$$

max: $a = 5/6$ $\quad b = 1/6$

(10) $\text{var}\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n \text{var}(x_i) + 2 \sum_{1 \le i < j \le n} \text{cov}(x_i, x_j)$

$$\text{cov}(x_i, x_j) = \rho(x_i x_j) \sqrt{\text{var}(x_i) \cdot \text{var}(x_j)} = 1/4$$

$$\Rightarrow n + \frac{n(n-1)}{4} = \frac{n(n+3)}{4}$$

4 4

(11) $P(X=0,Y=0)=0.24$ $Z,W=(0,0)$

$P(X=0,Y=1)=0.36$ $Z,W=(1,1)$

$P(X=1,Y=0)=0.16$ $Z,W=(1,1)$

$P(X=1,Y=1)=0.24$ $Z,W=(2,0)$

For bernoulli $var(X)=p-p^2$

$var(Z)=Var(X)+Var(Y)=0.48$

$P(W=1)=P(X=0,Y=1)+P(X=1,Y=0)=0.52$

$Var(W)=0.52(1-0.52^2)=0.2496$

$Cov(Z,W)=E[ZW]-E[Z]E[W]$

$E[Z]=E[X]+E[Y]=0.4+0.6$

$E[W]=1\times 0.52=0.52$

$E[Z,W]=0.36+0.16=0.52 \Rightarrow Cov(Z,W)=0$

$\rho(Z,W)=\frac{Cov(Z,W)}{\sqrt{Var(Z)\,Var(W)}}=0$

(14) (a) $\sum x f_N(x)=100\times 0.3=30$

(b) since independent $\Rightarrow 30/2=15$

(15) $T_1,T_2,\ldots,T_{500}$ be fail times

$m=Max(T_1,T_2,\ldots)$ $E[m]=?$

$F_m(m)=m^{500} \Rightarrow f_m(m)=500m^{499}$

$E[m]=\int_0^1 m.500m^{499}\,dm=500/501$ minutes